



Table of Fluorescent Proteins

This page is no longer being maintained. The interactive graph of fluorescent protein properties [<http://www.fpvis.org/>] and interactive graph of photoswitchable fluorescent protein properties [<http://www.fpvis.org/PSFP.html>] pages should be viewed instead.

Blue/UV Proteins

Protein	λ_{ex}	λ_{em}	Extinction coeff.	QY	Brightness	Aggregation	pKa	Source	Notes
TagBFP	402	457	52000	0.63	32.8	Monomer	2.7	Evrogen [http://www.evrogen.com]	aka mTagBFP; from [27]
mTagBFP2	399	454	50600	0.64	32.4	Monomer	2.7	[48]	
Azurite	383	450	26200	0.55	14.4	Monomer	5.0	[28]	
EBFP2	383	448	32000	0.56	18	Monomer	5.3	[29]	
mKalama1	385	456	36000	0.45	16	Monomer	5.5	[29]	
Sirius	355	424	15000	0.24	3.6	Monomer	<3.0	[37]	
Sapphire	399	511	29000	0.64	18.6	Monomer		[8]	aka H9-40
T-Sapphire	399	511	44000	0.6	26.4	Monomer		[12]	faster folding than Sapphire

Cyan Proteins

Protein	λ_{ex}	λ_{em}	Extinction coeff.	QY	Brightness	Aggregation	pKa	Source	Notes
ECFP	433	475	32500	0.4	13.0	Monomer	4.7		
Cerulean	433	475	43000	0.62	26.7	Monomer	4.7	[13]	improved ECFP
SCFP3A	433	474	30000	0.56	16.8	Monomer	<4.5	[39]	improved Cerulean
mTurquoise	434	474	30000	0.84	25.2	Monomer		[36]	improved SCFP3A
mTurquoise2	434	474	30000	0.93	27.9	Monomer	3.1	[47]	
monomeric Midoriishi-Cyan	470	496	22150	0.7	15.5	Monomer	7.0	MBL International [http://www.mblintl.com]	from [14]
TagCFP	458	480	37000	0.57	21.0	Monomer	4.7	Evrogen [http://www.evrogen.com]	
mTFP1	462	492	64000	0.85	54.0	Monomer	4.3	Allele Biotech [http://www.allelebiotech.com]	from [24]

Green Proteins

Protein	λ_{ex}	λ_{em}	Extinction coeff.	QY	Brightness	Aggregation	pKa	Source	Notes
EGFP	488	507	56000	0.6	33.6	Monomer			
Emerald	487	509	57500	0.68	37.3	Monomer		[8]	EGFP derivative; 5-fold brighter at 37° C
Superfolder GFP	485	510	83300	0.65	54.1	Monomer		[38]	Fast folding, highly stable
Monomeric Azami Green	492	505	55000	0.74	40.7	Monomer	5.8	MBL International [http://www.mblintl.com]	from [11]
TagGFP2	483	506	56500	0.6	33.9	Monomer	4.7	Evrogen	

								[http://www.evrogen.com]	
mUKG	483	499	60000	0.72	43.2	Monomer	5.2	[26]	
mWasabi	493	509	70000	0.80	56.0	Monomer	6.0	Allele Biotech [http://www.allelebiotech.com]	
Clover	505	515	111000	0.76	84.4	Monomer	6.1	[46]	
mNeonGreen	506	517	116000	0.80	92.8	Monomer	5.7	[49]	Allele Biotech [http://www.allelebiotech.com/mneongreen/]

Yellow Proteins

Protein	λ_{ex}	λ_{em}	Extinction coeff.	QY	Brightness	Aggregation	pKa	Source	Notes
EYFP	513	527	83400	0.61	50.9	Monomer			
Citrine	516	529	77000	0.76	58.5	Monomer	5.7	[2]	EYFP derivative; Less Cl, pH sensitive
Venus	515	528	92200	0.57	52.5	Monomer	6.0	[9]	EYFP derivative; 30-fold brighter at 37° C
SYFP2	515	527	101000	0.68	68.7	Monomer	6.0	[39]	improved Venus
TagYFP	508	524	64000	0.62	39.7	Monomer	5.5	Evrogen [http://www.evrogen.com]	

Orange Proteins

Protein	λ_{ex}	λ_{em}	Extinction coeff.	QY	Brightness	Aggregation	pKa	Source	Notes
Monomeric Kusabira-Orange	548	559	51600	0.6	31.0	Monomer	5.0	MBL International [http://www.mblintl.com]	from [14]
mKOκ	551	563	105000	0.61	64.0	Monomer	4.2	[26]	faster maturing than mKO
mKO2	551	565	63800	0.62	39.6	Monomer	5.5	MBL International [http://www.mblintl.com]	fast maturing
mOrange	548	562	71000	0.69	49.0	Monomer	6.5	[16]	~2.5 hr maturation time
mOrange2	549	565	58000	0.60	34.8	Monomer	6.5	[33]	~25x more photostable than mOrange

Red Proteins

Protein	λ_{ex}	λ_{em}	Extinction coeff.	QY	Brightness	Aggregation	pKa	Source	Notes
mRaspberry	598	625	86000	0.15	12.9	Monomer		[15]	~55 min maturation time
mCherry	587	610	72000	0.22	15.8	Monomer	<4.5	[16]	~15 min maturation time
mStrawberry	574	596	90000	0.29	26.1	Monomer	<4.5	[16]	~50 min maturation time
mTangerine	568	585	38000	0.3	11.4	Monomer	5.7	[16]	
tdTomato	554	581	138000	0.69	95.2	Monomer	4.7	[16]	~1 hr maturation time
TagRFP	555	584	100000	0.48	49.0	Monomer	3.8	Evrogen [http://www.evrogen.com]	From [20]
TagRFP-T	555	584	81000	0.41	33.2	Monomer	4.6	[33]	~10x more photostable than TagRFP
mApple	568	592	75000	0.49	36.7	Monomer	6.5	[33]	
mRuby	558	605	112000	0.35	39.2	Monomer	4.4	[35]	
mRuby2	559	600	113000	0.38	43	Monomer	5.3	[46]	

Far-Red Proteins

Protein	λ_{ex}	λ_{em}	Extinction coeff.	QY	Brightness	Aggregation	pKa	Source	Notes
mPlum	590	649		0.1		Monomer		[15]	~100 min maturation time
HcRed-Tandem	590	637	160000	0.04	6.4	Monomer			Tandem dimer; functional monomer
mKate2	588	633	62500	0.40	25	Monomer	5.4	Evrogen [http://www.evrogen.com]	
mNeptune	600	650	67000	0.20	13.4	Monomer	5.4	[34]	
NirFP	605	670	15700	0.06	0.9	Dimer	4.5	Evrogen [http://www.evrogen.com]	

Near-IR Proteins

TagRFP657	611	657	34000	0.10	3.4	Monomer	5.0	[30]	
IFP1.4	684	708	102000	0.077	7.8	Monomer	4.6	[31]	Bacterial phytochrome; requires biliverdin cofactor for fluorescence
iRFP	690	713	105000	0.059	6.2	Dimer	4.0	[43]	Bacterial phytochrome; requires biliverdin cofactor for fluorescence

Long Stokes Shift Proteins

Protein	λ_{ex}	λ_{em}	Extinction coeff.	QY	Brightness	Aggregation	pKa	Source	Notes
mKeima Red	440	620	14400	0.24	3.5	Monomer	6.5	MBL International [http://www.mblintl.com]	From [22]
LSS-mKate1	463	624	31200	0.08	2.5	Monomer	3.2	[32]	
LSS-mKate2	460	605	26000	0.17	4.4	Monomer	2.7	[32]	
mBeRFP	446	611	65000	0.27	17.6	Monomer	5.6	[50]	

Photoactivatable Proteins

Protein	λ_{ex}	λ_{em}	Extinction coeff.	QY	Brightness	Aggregation	pKa	Source	Notes
PA-GFP	504	517	17400	0.79	13.7	Monomer		[10]	
PAmCherry1	564	595	18000	0.46	8.3	Monomer	6.3	[41]	
PATagRFP	562	595	66000	0.38	25.1	Monomer	5.3	[40]	

Photoconvertible Proteins

Protein	λ_{ex}	λ_{em}	Extinction coeff.	QY	Brightness	Aggregation	pKa	Source	Notes
Kaede (green)	508	518	98800	0.88	86.9	Tetramer	5.6	MBL International [http://www.mblintl.com]	
Kaede (red)	572	580	60400	0.33	19.9	Tetramer	5.6	MBL International [http://www.mblintl.com]	after photoconversion
KikGR1 (green)	507	517	53700	0.7	37.6	Tetramer	7.8	MBL International [http://www.mblintl.com]	from [23]
KikGR1 (red)	583	593	35100	0.65	22.8	Tetramer	5.5	MBL International [http://www.mblintl.com]	after photoconversion; from [23]
PS-CFP2	400	468	43000	0.2	8.6	Monomer		Evrogen [http://www.evrogen.com]	Before photoconversion
PS-CFP2	490	511	47000	0.23	10.8	Monomer		Evrogen	After photoconversion

								[http://www.evrogen.com]	
mEos2 (green)	506	519	56000	0.84	47.0	Monomer	5.6	[42]	
mEos2 (red)	573	584	46000	0.66	30.4	Monomer	6.4	[42]	After photoconversion
mEos3.2 (green)	507	516	63400	0.70	53	Monomer	5.4	[45]	Better folding, more monomeric than mEos2
mEos3.2 (red)	572	580	32200	0.55	18	Monomer	5.8	[45]	After photoconversion
PSmOrange	548	565	113300	0.51	57.8	Monomer	6.2	[44]	Before photoconversion
PSmOrange	634	662	32700	0.28	9.2	Monomer	5.6	[44]	After photoconversion

Photoswitchable Proteins

Protein	λ_{ex}	λ_{em}	Extinction coeff.	QY	Brightness	Aggregation	pKa	Source	Notes
Dronpa	503	518	95000	0.85	80.7	Monomer		MBL International [http://www.mblint.com]	From [19]; Photoactivated form

Notes:

This is not a complete list of existing fluorescent proteins. In general I have included only those proteins that are commercially available or that are well characterized and represent a clear improvement over existing proteins. All values were taken from the manufacturers data or the indicated papers. I cannot vouch for their reliability. Brightness is the product of extinction coefficient and quantum yield, divided by 1000.

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